Project 1: Numerical Solutions to Elliptic PDEs

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Write a concise abstract (100–200 words). State the PDEs solved, numerical methods used (e.g., second-order finite differences, conjugate gradient, time stepping), and key outcomes (accuracy, convergence, settling time to steady state, etc.)

I. INTRODUCTION

Briefly motivate the problem (2D heat equation with source; steady Poisson limit), and summarize what you implemented and what you will report (mesh, solver, error, and results figures)[1].

II. GOVERNING EQUATIONS AND BOUNDARY CONDITIONS

State the PDE(s) and boundary conditions clearly. For example, the 2D heat equation with source,

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \kappa(x, y), \tag{1}$$

and its steady-state Poisson equation,

$$\alpha \left(\frac{\partial^2 \bar{u}}{\partial x^2} + \frac{\partial^2 \bar{u}}{\partial y^2} \right) = -\kappa(x, y). \tag{2}$$

List the mixed boundary conditions on each side of the square domain.

III. NUMERICAL METHOD

III.A. Grid and Discretization

Describe the uniform grid, h_x, h_y, indexing, and the second-order central stencil. Include the 5-point Laplacian formula.

III.B. Linear System and Solver (Poisson)

Describe matrix form $A\mathbf{u} = \mathbf{b}$, sparsity, and the conjugate gradient method (tolerance, max iterations, stopping criterion).

III.C. Time Integration (Heat Equation)

State your time-marching method (e.g., forward Euler / backward Euler / Crank–Nicolson), time step Δt, and the stopping criterion:

$$\frac{\|u(x, y, t) - \bar{u}(x, y)\|_2}{\|\bar{u}(x, y)\|_2} < 10^{-4}. \tag{3}$$

IV. RESULTS

IV.A. Poisson Steady-State Solution

Show the contour/heatmap of $\bar{u}(x, y)$ and discuss qualitative behavior.

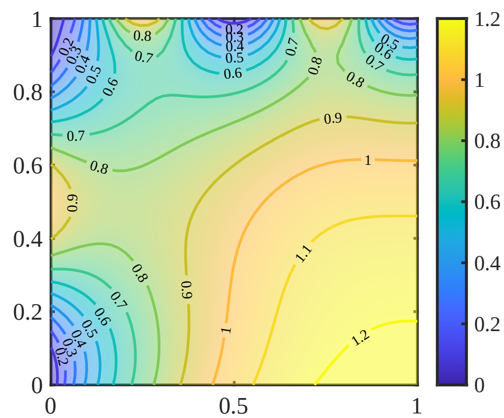

FIG. 1 Steady-state solution $\bar{u}(x, y)$ obtained by solving the Poisson equation using second-order finite differences and conjugate gradient.

IV.B. Transient Heat Equation and Settling Time

Show snapshots or a convergence plot versus time, and report the settling time t^*.

IV.C. Grid Sensitivity (Optional but Recommended)

Briefly show that results are mesh-independent (e.g., 21×21, 31×31).

V. DISCUSSION

Interpret results: solver performance, convergence trends, stability constraints (if explicit), and any numerical issues (boundary treatment, corners, conditioning, etc.).

VI. CONCLUSION

Summarize the main takeaways in 4–6 sentences. Clearly state the final reported settling time and the steady-state match.

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[1] Y. Saad, *Iterative Methods for Sparse Linear Systems* (SIAM, 2003).